

Definition of Pharmacology

Pharmacology is the branch of science that deals with the study of drugs, their origin, composition, pharmacokinetics, pharmacodynamics, therapeutic uses, and adverse effects. It examines how drugs interact with biological systems to produce effects and their mechanisms of action.

The field of pharmacology is divided into two main areas:

Pharmacokinetics: The study of how the body absorbs, distributes, metabolizes, and excretes drugs.

Pharmacodynamics: The study of how drugs interact with their biological targets to produce therapeutic or toxic effects.

Principles of General Pharmacology

1. Drug Absorption

- Refers to the process by which a drug enters the bloodstream from its site of administration.

2. Drug Distribution

- Once absorbed, the drug is distributed throughout the body to reach its site of action.

3. Drug Metabolism

- Metabolism is the biochemical modification of drugs, primarily in the liver.
- The process often transforms drugs into more water-soluble compounds for easier excretion.

4. Drug Excretion

- The removal of drugs and their metabolites from the body.

5. Mechanism of Action

- Drugs exert their effects by binding to specific molecular targets like receptors, enzymes, or ion channels.

6. Dose-Response Relationship

- Describes the relationship between the dose of a drug and its pharmacological effect.

7. Therapeutic Index

- Ratio of the toxic dose to the effective dose (TD50/ED50).

8. Adverse Drug Reactions (ADRs)

- Unintended, harmful effects caused by drugs.

9. Drug Interactions

- Occur when one drug affects the action of another.

10. Individual Variations

- Drug response varies among individuals due to:
- Genetic factors (pharmacogenomics).
- Age and gender.
- Physiological and pathological conditions (e.g., kidney or liver dysfunction).

11. Routes of Drug Administration

- The method of drug delivery influences its onset, intensity, and duration of action.
- Common routes:
- Enteral: Oral, sublingual, rectal.
- Parenteral: Intravenous, intramuscular, subcutaneous.
- Topical: Skin, mucous membranes.
- Inhalation: Respiratory tract.

Definition of Drug

A drug is any chemical substance that, when administered to a living organism, produces a biological effect. Drugs are used for diagnosis, prevention, treatment, or cure of diseases, as well as for modifying physiological functions.

Examples: Antibiotics, analgesics, antipyretics, and hormones.

Drug Dosage Forms

Drug dosage forms are the physical forms in which drugs are manufactured and administered to ensure accurate dosing, stability, and patient acceptability. These can be broadly classified into solid, liquid, and semisolid forms.

1. Solid Dosage Forms

- Tablets: Compressed powder containing active drug and excipients. Can be chewable, effervescent, or coated.
- Capsules: Drugs enclosed in a gelatin shell for controlled release.
- Powders: Fine particles of the drug, often dissolved before administration.
- Granules: Larger, coarse particles that are often reconstituted before use.

2. Liquid Dosage Forms

- Solutions: Homogeneous mixtures of the drug in water or another solvent.
- Syrups: Sweetened, flavored liquid formulations, mainly for pediatric use.
- Elixirs: Clear, sweetened hydro-alcoholic liquids.
- Suspensions: Insoluble drugs dispersed in a liquid medium.
- Emulsions: Mixtures of two immiscible liquids stabilized by an emulsifier.

3. Semi-Solid Dosage Forms

- Ointments: Oil-based preparations for external application.
- Creams: Semi-solid emulsions for topical use.
- Gels: Transparent, jelly-like formulations for localized treatment.
- Pastes: Thick preparations containing a high proportion of solids.

4. Miscellaneous Dosage Forms

- Inhalers: Devices delivering medication directly to the lungs.
- Transdermal Patches: Adhesive patches for systemic drug delivery through the skin.
- Suppositories: Solid dosage forms inserted into body cavities (rectum, vagina).
- Injectables: Sterile solutions or suspensions for parenteral use.

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Routes of Drug Administration

The route of administration refers to the path through which a drug is delivered to achieve its therapeutic effect. The choice depends on the drug's properties, the desired onset of action, and the patient's condition.

1. Enteral Routes

Oral (PO): The most common and convenient route.

Advantages: Non-invasive, easy to administer, suitable for self-administration.

Disadvantages: Slow onset, subject to first-pass metabolism.

Sublingual: Drug placed under the tongue for rapid absorption.

Example: Nitroglycerin for angina.

Rectal: Drug administered via the rectum.

Uses: For patients unable to take drugs orally (e.g., vomiting, unconscious).

2. Parenteral Routes (Bypassing the gastrointestinal tract)

Intravenous (IV): Direct injection into the bloodstream.

Advantages: Immediate effect, 100% bioavailability.

Disadvantages: Invasive, requires skilled administration.

Intramuscular (IM): Injection into muscle tissue.

Example: Vaccines, antibiotics.

Subcutaneous (SC): Injection under the skin.

Example: Insulin, heparin.

Intradermal (ID): Injection into the dermis layer.

Example: Tuberculin test.

Intra-articular: Injection into a joint cavity.

Example: Corticosteroids for arthritis.

3. Topical Routes

Drug applied directly to the skin or mucous membranes for local effects.

Examples: Creams, ointments, patches.

4. Inhalation

Drugs administered as gases, aerosols, or vapors for respiratory or systemic effects.

Examples: Asthma inhalers (bronchodilators), anesthetic gases.

5. Transdermal

Drugs delivered through the skin for systemic absorption.

Example: Nicotine patches, fentanyl patches.

6. Specialized Routes

Intranasal: Drugs administered through the nasal cavity.

Example: Nasal sprays for allergies.

Ophthalmic: Drugs applied to the eye.

Example: Eye drops for glaucoma.

Otic: Drugs administered into the ear.

Example: Ear drops for infections.

Pharmacokinetics (ADME)

Pharmacokinetics studies the movement of drugs within the body, often summarized by the acronym ADME: Absorption, Distribution, Metabolism, and Excretion. It answers the question, "What does the body do to the drug?"

1. Absorption

Definition: The process by which a drug enters the bloodstream from its site of administration.

Factors affecting absorption:

Route of administration: Oral, intravenous, sublingual, etc.

Drug formulation: Tablets, capsules, liquids.

Solubility: Water-soluble or lipid-soluble drugs.

pH: Affects ionization and permeability.

Presence of food: May delay or enhance absorption.

Example: Oral drugs like paracetamol are absorbed through the gastrointestinal tract.

2. Distribution

Definition: The process by which a drug is transported from the bloodstream to various tissues and organs.

Factors influencing distribution:

Blood flow to tissues.

Binding of drugs to plasma proteins (e.g., albumin).

Permeability of cell membranes (e.g., blood-brain barrier).

Volume of distribution (Vd): A measure of how extensively a drug is distributed in body tissues.

Example: Lipophilic drugs, like diazepam, accumulate in fat tissues.

3. Metabolism

Definition: The biochemical transformation of drugs into active or inactive metabolites, primarily occurring in the liver.

Phases of metabolism: स्वास्थ्य रक्षणं, आतुरस्य विकार प्रशमनं च ।

Phase I: Functionalization reactions (oxidation, reduction, hydrolysis).

Example: Conversion of codeine to morphine.

Phase II: Conjugation reactions (glucuronidation, sulfation).

Example: Conjugation of paracetamol for elimination.

First-pass metabolism: Drugs metabolized significantly in the liver before reaching systemic circulation.

Example: Nitroglycerin is rapidly metabolized if taken orally.

4. Excretion

Definition: The elimination of drugs and their metabolites from the body.

Routes of excretion:

Renal excretion (urine): Most common route (e.g., antibiotics like amoxicillin).

Hepatic excretion (bile and feces): For lipophilic drugs.

Other routes: Sweat, saliva, exhaled air.

Example: Renal clearance of aminoglycosides like gentamicin.

Pharmacodynamics

Pharmacodynamics studies the biological and physiological effects of drugs on the body and their mechanisms of action. It answers the question, "What does the drug do to the body?"

1. Mechanism of Action

Drugs produce effects by interacting with biological targets:

Receptors: Proteins that drugs bind to, initiating a response.

Agonists: Activate receptors to produce a response (e.g., salbutamol for asthma).

Antagonists: Block receptors and inhibit a response (e.g., propranolol for hypertension).

Enzymes: Drugs may inhibit or activate enzymes (e.g., aspirin inhibits cyclooxygenase enzymes).

Ion channels: Drugs modulate ion flow across membranes (e.g., calcium channel blockers for hypertension).

2. Dose-Response Relationship

Definition: The relationship between the dose of a drug and the magnitude of its effect.

Key terms:

Efficacy: The maximum effect a drug can produce.

Potency: The dose required to produce a given effect.

Therapeutic window: The range of doses between the effective dose (ED50) and the toxic dose (TD50).

3. Types of Drug Effects

Therapeutic Effects: Desired and beneficial effects (e.g., pain relief with analgesics).

Adverse Effects: Unintended, harmful effects.

Dose-dependent (predictable): Overdose or exaggerated response.

Idiosyncratic (unpredictable): Genetic or immune-related responses.

Toxic Effects: Severe harmful effects at high doses (e.g., liver damage from paracetamol overdose).

4. Drug-Receptor Interactions

Affinity: The ability of a drug to bind to a receptor.

Intrinsic activity: The ability of a drug to produce a biological response after binding.

Examples:

Morphine (agonist) binds to opioid receptors for pain relief.

Naloxone (antagonist) reverses opioid overdose.

5. Factors Influencing Drug Response

Individual variability:

Age, gender, weight.

Genetic factors (pharmacogenomics).

Physiological conditions (e.g., kidney or liver disease).

Drug tolerance: Reduced response to a drug after repeated use.

Drug interactions: Synergism (enhanced effect) or antagonism (reduced effect) with other drugs.

1. Drug Dose

A drug dose refers to the quantity of a drug administered at one time to achieve the desired therapeutic effect without causing toxicity.

Types of Doses:

Therapeutic dose: The amount of drug required to produce the desired effect.

Loading dose: A higher initial dose given to quickly achieve therapeutic levels.

Maintenance dose: A smaller dose given to maintain drug concentration within the therapeutic range.

Toxic dose: The amount of drug that produces harmful effects.

Lethal dose: The dose of a drug that can cause death.

Factors Affecting Dose:

Age, weight, and gender.

Disease state (e.g., liver or kidney impairment).

Route of administration (oral, IV, etc.).

Drug formulation and bioavailability.

2. Principles of Drug Action

The principles of drug action describe how drugs produce their effects in the body.

Key Principles:

Selective Action:

Drugs act on specific cells, tissues, or receptors.

Example: Beta-blockers selectively act on beta-adrenergic receptors.

Dose-Response Relationship:

The response to a drug depends on the dose given. Higher doses generally produce greater effects, up to a maximum limit.

Reversible Action:

Many drug effects are reversible, allowing the body to return to its normal state once the drug is metabolized or excreted.

Example: Insulin temporarily lowers blood sugar levels.

Agonism and Antagonism:

Agonists: Activate receptors to produce effects.

Antagonists: Block receptors and prevent other substances from binding.

Time of Action:

Drugs act over different time frames:

Onset of action: When the effect begins.

Peak effect: Maximum therapeutic effect.

Duration of action: How long the effect lasts.

Therapeutic Index:

The ratio of a drug's toxic dose to its effective dose (TD50/ED50), indicating its safety margin.

Mechanism of Drug Action

The mechanism of drug action explains how a drug interacts with the body to produce its effects.

1. Interaction with Receptors:

Drugs bind to specific receptors on cell membranes or within cells.

Agonists: Activate receptors to produce a biological response (e.g., morphine activates opioid receptors).

Antagonists: Block receptors to prevent a response (e.g., naloxone blocks opioid receptors).

Partial agonists: Activate receptors but produce a weaker response compared to full agonists.

2. Enzyme Modulation:

Drugs can inhibit or activate enzymes.

Example: Aspirin inhibits cyclooxygenase (COX) enzymes to reduce inflammation.

3. Ion Channel Modulation:

Drugs regulate ion flow through cell membrane channels.

Example: Calcium channel blockers (e.g., amlodipine) reduce calcium influx to lower blood pressure.

4. Interaction with Transporters:

Drugs interfere with transport proteins that move substances across membranes.

Example: Selective serotonin reuptake inhibitors (SSRIs) block serotonin reuptake to treat depression.

5. Chemical Action:

Some drugs act chemically without involving receptors or enzymes.

Example: Antacids neutralize stomach acid through chemical reactions.

6. Non-Specific Actions:

Some drugs act on multiple pathways or targets.

Example: General anesthetics alter membrane fluidity to depress the central nervous system.

Bioavailability

Bioavailability refers to the proportion of an administered drug that reaches systemic circulation in its active form.

Factors Affecting Bioavailability:

Route of Administration:

Intravenous (IV) administration has 100% bioavailability.

Oral drugs may have reduced bioavailability due to first-pass metabolism in the liver.

Drug Formulation:

Solubility, particle size, and excipients affect absorption.

First-Pass Metabolism:

Drugs metabolized in the liver before entering systemic circulation have reduced bioavailability.

Example: Propranolol has low oral bioavailability due to extensive first-pass metabolism.

Gastrointestinal Factors:

Presence of food, pH, and enzyme activity in the gastrointestinal tract influence absorption.

Example: High-fat meals increase the absorption of lipophilic drugs.

Protein Binding:

Drugs bound to plasma proteins (e.g., albumin) are pharmacologically inactive until released.

Significance of Bioavailability:

Determines the dosage required to achieve therapeutic drug levels.

Helps in comparing different formulations or routes of administration of the same drug.

1. Anaesthetics

Definition: Drugs that cause a temporary loss of sensation or consciousness, used during surgical or diagnostic procedures.

Detailed Mode of Action:

General Anaesthetics: These drugs suppress brain activity by acting on specific receptors in the central nervous system (CNS). For example:

They enhance the effect of inhibitory neurotransmitters like GABA, which slows down brain activity.

They also block excitatory neurotransmitters like glutamate, reducing brain signals.

Local Anaesthetics: They block sodium ion channels in nerve cells. This prevents the nerves from transmitting pain signals to the brain.

Types and Examples:

General Anaesthetics: Halothane, Propofol, Ketamine.

Local Anaesthetics: Lidocaine, Bupivacaine.

2. Sedative-Hypnotic Drugs

Definition: Drugs that calm the mind (sedation) or induce sleep (hypnosis).

Detailed Mode of Action:

These drugs work by enhancing the action of GABA, the brain's main inhibitory neurotransmitter. GABA reduces the activity of nerve cells, making the brain less excitable.

Sedatives act mildly, calming the brain, while hypnotics produce stronger effects, leading to sleep.

Some of these drugs act directly on GABA-A receptors, increasing the flow of chloride ions into the cells, which hyperpolarizes the neurons and makes them less likely to fire.

Types and Examples:

Benzodiazepines: Diazepam, Lorazepam.

Non-benzodiazepines: Zolpidem, Zopiclone.

Barbiturates: Phenobarbital.

3. Antiepileptic Drugs (AEDs)

Definition: Drugs used to prevent or control seizures.

Detailed Mode of Action:

Seizures happen due to abnormal electrical activity in the brain. Antiepileptic drugs control this by:

Blocking Sodium Channels: They reduce the excessive firing of neurons by stabilizing sodium channels in their inactive state, so the neuron cannot keep firing repeatedly.

Blocking Calcium Channels: They reduce the release of neurotransmitters involved in abnormal electrical signaling.

Enhancing GABA Activity: Some AEDs increase the effect of GABA, which calms overactive neurons.

Types and Examples:

Sodium Channel Blockers: Phenytoin, Carbamazepine.

Calcium Channel Blockers: Ethosuximide.

GABA Enhancers: Valproic acid, Clonazepam.

4. Antiparkinsonian Drugs

Definition: Drugs used to treat symptoms of Parkinson’s disease by restoring dopamine balance.

Detailed Mode of Action:

Parkinson’s disease is caused by low levels of dopamine in the brain due to the degeneration of dopamine-producing neurons. These drugs work by:

Dopamine Precursors: Levodopa converts into dopamine in the brain, replenishing the lost dopamine.

Carbidopa prevents the breakdown of levodopa in the blood, so more reaches the brain.

Dopamine Agonists: They directly stimulate dopamine receptors in the brain, mimicking the effects of dopamine.

MAO-B Inhibitors: These block the enzyme monoamine oxidase-B (MAO-B), which breaks down dopamine in the brain, increasing dopamine levels.

Types and Examples:

Dopamine Precursors: Levodopa + Carbidopa.

Dopamine Agonists: Pramipexole, Ropinirole.

MAO-B Inhibitors: Selegiline.

5) Antidepressants

Definition: Drugs that treat depression by improving the balance of neurotransmitters like serotonin and norepinephrine.

Detailed Mode of Action:

Depression is linked to low levels of neurotransmitters like serotonin and norepinephrine in the brain. These drugs work by:

SSRIs: They block the reuptake of serotonin back into the nerve endings, keeping more serotonin available in the brain.

SNRIs: They block the reuptake of both serotonin and norepinephrine, increasing their levels in the brain.

TCAs: Similar to SNRIs, but they also affect other receptors, leading to more side effects.

Types and Examples:

SSRIs: Fluoxetine, Sertraline.

SNRIs: Venlafaxine, Duloxetine.

TCAs: Amitriptyline, Imipramine.

6. Antianxiety Drugs

Definition: Drugs that reduce anxiety and induce relaxation.

Detailed Mode of Action:

Anxiety occurs due to overactivity in certain brain circuits. These drugs work by:

Enhancing GABA activity, which calms the brain by reducing the activity of overactive neurons.

Some drugs like beta-blockers reduce the physical symptoms of anxiety (e.g., fast heart rate) by blocking the effects of adrenaline.

Types and Examples:

Benzodiazepines: Short-term relief, e.g., Alprazolam, Diazepam.

Non-Benzodiazepines: Safer for long-term use, e.g., Buspirone.

Beta-Blockers: Control physical symptoms, e.g., Propranolol.

7. Opioid Analgesics

Definition: Drugs that relieve severe pain by acting on opioid receptors in the brain and spinal cord.

Detailed Mode of Action:

Opioids bind to mu, kappa, and delta opioid receptors in the brain and spinal cord.

This blocks the transmission of pain signals to the brain.

They also increase dopamine release, which creates a sense of euphoria.

In the spinal cord, opioids inhibit the release of neurotransmitters that send pain signals to the brain.

Types and Examples:

Natural Opioids: Morphine, Codeine.

Synthetic Opioids: Fentanyl, Tramadol.

DEFINE, DESCRIBE MODE OF ACTION & DISCUSS TYPES WITH EXAMPLES OF FOLLOWING DRUGS ACTING ON PERIPHERAL (SOMATIC) NERVOUS SYSTEM WITH EXAMPLES OF SKELETAL MUSCLE RELAXANTS AND LOCAL ANAESTHETICS DRUGS

1. Skeletal Muscle Relaxants

Definition:

Drugs that reduce muscle tone and relax skeletal muscles. They are used in conditions like muscle spasms, spasticity, or during surgery to facilitate procedures.

Detailed Mode of Action:

These drugs act at the neuromuscular junction (where nerves connect with muscles) or in the spinal cord. They work by either:

Blocking Nerve Signals: Prevent the transmission of signals from nerves to muscles by interfering with acetylcholine (a neurotransmitter required for muscle contraction).

Central Action: Reduce signals from the brain and spinal cord to relax muscles.

Types and Examples:

Peripherally Acting Muscle Relaxants:

Block acetylcholine at the neuromuscular junction.

Examples:

Non-depolarizing agents: Block acetylcholine receptors (e.g., Atracurium, Vecuronium).

Depolarizing agents: Mimic acetylcholine initially but then block its action (e.g., Succinylcholine).

Centrally Acting Muscle Relaxants:

Act on the central nervous system to reduce muscle spasticity.

Examples: Baclofen, Diazepam.

2. Local Anaesthetics

Definition:

Drugs that temporarily block pain and sensation in a specific area without affecting consciousness.

Detailed Mode of Action:

Local anaesthetics block voltage-gated sodium channels in nerve cells.

This prevents the entry of sodium ions into the nerve, stopping the conduction of electrical signals.

As a result, the nerve cannot send pain signals to the brain.

The effects are reversible, meaning the nerve function is restored after the drug wears off.

Types and Examples:

Amide-Type Local Anaesthetics:

Longer duration of action due to slower metabolism in the liver.

Examples: Lidocaine, Bupivacaine, Ropivacaine.

Ester-Type Local Anaesthetics:

Shorter duration of action as they are metabolized quickly by enzymes in the blood.

Examples: Procaine, Tetracaine.

Define, Describe mode of action & Discuss types with examples of following drugs acting on Autacoids and Related of Nonsteroidal Antiinflammatory (NSAIDs), Antipyretic and Analgesics Drug

1. Nonsteroidal Anti-inflammatory Drugs (NSAIDs)

Definition:

Drugs that reduce inflammation, pain, and fever by blocking certain chemical messengers in the body called prostaglandins.

Detailed Mode of Action:

Prostaglandins are chemicals produced in the body during injury or inflammation that cause pain, swelling, and fever.

NSAIDs inhibit the enzyme cyclooxygenase (COX), which is responsible for prostaglandin synthesis. There are two types of COX enzymes:

COX-1: Produces protective prostaglandins that protect the stomach lining and maintain kidney function.

COX-2: Produces prostaglandins responsible for inflammation and pain.

By inhibiting COX enzymes, NSAIDs reduce inflammation, pain, and fever, but blocking COX-1 can lead to side effects like stomach irritation.

Types and Examples:

Non-selective COX Inhibitors: Inhibit both COX-1 and COX-2.

Examples: Aspirin, Ibuprofen, Diclofenac.

Selective COX-2 Inhibitors: Specifically inhibit COX-2, with fewer stomach-related side effects.

Examples: Celecoxib, Etoricoxib.

Salicylates: A subtype of NSAIDs derived from salicylic acid.

Example: Aspirin (also has antiplatelet action).

2. Antipyretic Drugs

Definition:

Drugs that reduce fever by acting on the thermoregulatory center in the brain to lower body temperature.

Detailed Mode of Action:

Fever occurs when the hypothalamus (body's thermostat) is reset to a higher temperature in response to pyrogens (substances that induce fever).

Antipyretics work by:

Blocking prostaglandin synthesis in the hypothalamus by inhibiting COX enzymes.

This normalizes the hypothalamic set point and reduces body temperature.

They do not affect normal body temperature in the absence of fever.

Examples:

Paracetamol (Acetaminophen): Widely used for fever with minimal anti-inflammatory action.

Ibuprofen: Acts as both an antipyretic and an anti-inflammatory.

3. Analgesics

Definition:

Drugs that relieve pain without causing loss of consciousness.

Detailed Mode of Action:

Pain is caused by the release of prostaglandins, bradykinin, and other chemicals during tissue injury or inflammation.

Analgesics block the production or action of these pain-causing chemicals:

Non-opioid Analgesics:

Inhibit COX enzymes, reducing prostaglandins responsible for pain and inflammation.

Examples: Paracetamol, NSAIDs (Ibuprofen, Diclofenac).

Opioid Analgesics(not autacoid-related but relevant for comparison):

Act on opioid receptors in the brain and spinal cord to block pain signals.

Types and Examples:

Non-opioid Analgesics:

Examples: Paracetamol (for mild to moderate pain), NSAIDs like Ibuprofen.

Combination Drugs:

Example: Paracetamol + Ibuprofen (for enhanced pain relief).

1. Bronchodilators

Definition:

Drugs that relax the muscles of the airways, widen the bronchi, and improve airflow, commonly used in conditions like asthma and chronic obstructive pulmonary disease (COPD).

Detailed Mode of Action:

Bronchodilators target the smooth muscles of the respiratory airways, reducing bronchoconstriction.

They work by:

Beta-2 Adrenergic Agonists: Stimulate beta-2 receptors on airway smooth muscles, increasing cyclic AMP (cAMP) levels. This relaxes the muscles and opens the airways.

Anticholinergics: Block muscarinic receptors, preventing acetylcholine from causing airway narrowing.

Methylxanthines: Inhibit phosphodiesterase enzymes, increasing cAMP and relaxing airway smooth muscles.

Types and Examples:

Beta-2 Adrenergic Agonists:

Short-acting: Salbutamol (Albuterol), Terbutaline.

Long-acting: Salmeterol, Formoterol.

Anticholinergics:

Short-acting: Ipratropium.

Long-acting: Tiotropium.

Methylxanthines:

Examples: Theophylline, Aminophylline.

2. Aerosols/Inhalants

Definition:

Drugs administered directly into the lungs via inhalation for local effects in the respiratory system.

Detailed Mode of Action:

Aerosols and inhalants deliver medication directly to the airways, reducing systemic side effects and providing faster relief.

These drugs are often used for asthma and COPD, and they act on various mechanisms depending on the active ingredient:

Bronchodilation: Beta-2 agonists or anticholinergics relax airway muscles.

Anti-inflammatory Effects: Corticosteroids reduce inflammation in the airways.

Mucolytic Effects: Thin mucus for easier clearance.

Types and Examples:

Bronchodilators via Inhalation: Salbutamol (short-acting), Salmeterol (long-acting).

Corticosteroids: Budesonide, Fluticasone.

Combination Inhalers: Salmeterol + Fluticasone, Budesonide + Formoterol.

3. Expectorants

Definition:

Drugs that help loosen and thin mucus in the airways, making it easier to cough up and clear from the respiratory tract.

Detailed Mode of Action:

Expectorants increase the secretion of fluid in the respiratory tract, which reduces the thickness and stickiness of mucus.

They may:

Stimulate Mucus Secretion: Increase the volume of respiratory secretions.

Reduce Mucus Viscosity: Thin the mucus, making it easier to expel.

Examples:

Guaifenesin: Most commonly used expectorant to thin mucus.

Potassium Iodide: Increases respiratory tract secretions.

Ammonium Chloride: Used in some formulations to aid mucus clearance.

4. Antitussives (Cough Suppressants)

Definition:

Drugs that suppress or reduce the urge to cough. They are used in cases of dry, non-productive cough.

Detailed Mode of Action:

Central Action: Act on the cough center in the brain (medulla) to suppress the cough reflex.

Peripheral Action: Reduce irritation in the throat or airways that trigger the cough reflex.

Types and Examples:

- Centrally Acting Antitussives:

Opioid-Based: Codeine, Hydrocodone (suppress the brain’s cough center).

Non-Opioid: Dextromethorphan (commonly used, fewer side effects).

- Peripherally Acting Antitussives:

Soothing agents like menthol, which reduce throat irritation.

Examples: Lozenges, Syrups containing menthol or honey.

Define, Describe mode of action & Discuss types with examples of following drugs acting on Cardiovascular Drugs as Antihypertensive and Antianginal Drugs

1. Antihypertensive Drugs

Definition:

Drugs used to lower high blood pressure (hypertension) to prevent complications like heart attack, stroke, or kidney damage.

Detailed Mode of Action:

Hypertension occurs due to increased resistance in blood vessels, high blood volume, or overactive signals from the nervous system. Antihypertensive drugs work by:

- Reducing Blood Vessel Constriction: Relax the blood vessels, lowering resistance to blood flow.
- Decreasing Blood Volume: Promote the excretion of sodium and water from the body, reducing blood pressure.
- Lowering Heart Rate and Cardiac Output: Reduce the workload of the heart.

Types and Examples:

- Diuretics:** Help the kidneys remove excess sodium and water from the body.

Examples: Hydrochlorothiazide, Furosemide.

- ACE Inhibitors:** Block the conversion of angiotensin I to angiotensin II, a hormone that constricts blood vessels.

Examples: Enalapril, Lisinopril.

- Angiotensin II Receptor Blockers (ARBs):** Block the action of angiotensin II directly at its receptor.

Examples: Losartan, Valsartan.

- Beta-Blockers:** Reduce heart rate and contractility, lowering blood pressure.

Examples: Atenolol, Metoprolol.

- Calcium Channel Blockers (CCBs):** Block calcium entry into smooth muscle cells, relaxing blood vessels and reducing heart workload.

Examples: Amlodipine, Verapamil.

- Alpha-Blockers:** Relax blood vessels by blocking alpha receptors.

Examples: Prazosin, Doxazosin.

2. Antianginal Drugs

Definition:

Drugs that relieve chest pain (angina) caused by reduced blood flow and oxygen to the heart muscle.

Detailed Mode of Action:

Angina occurs when the heart muscle doesn't get enough oxygen due to reduced blood flow (e.g., during exertion). Antianginal drugs work by:

- Increasing Oxygen Supply: Dilate coronary arteries to improve blood flow to the heart muscle.
- Decreasing Oxygen Demand: Reduce the heart's workload by lowering heart rate, contractility, or blood pressure.

Types and Examples:

- Nitrates:

Relax blood vessels (vasodilation) by increasing nitric oxide levels, reducing the heart's oxygen demand and improving blood flow.

Examples: Nitroglycerin, Isosorbide Mononitrate.

- Beta-Blockers:

Decrease heart rate and contractility, reducing oxygen demand.

Examples: Atenolol, Propranolol.

- Calcium Channel Blockers (CCBs):

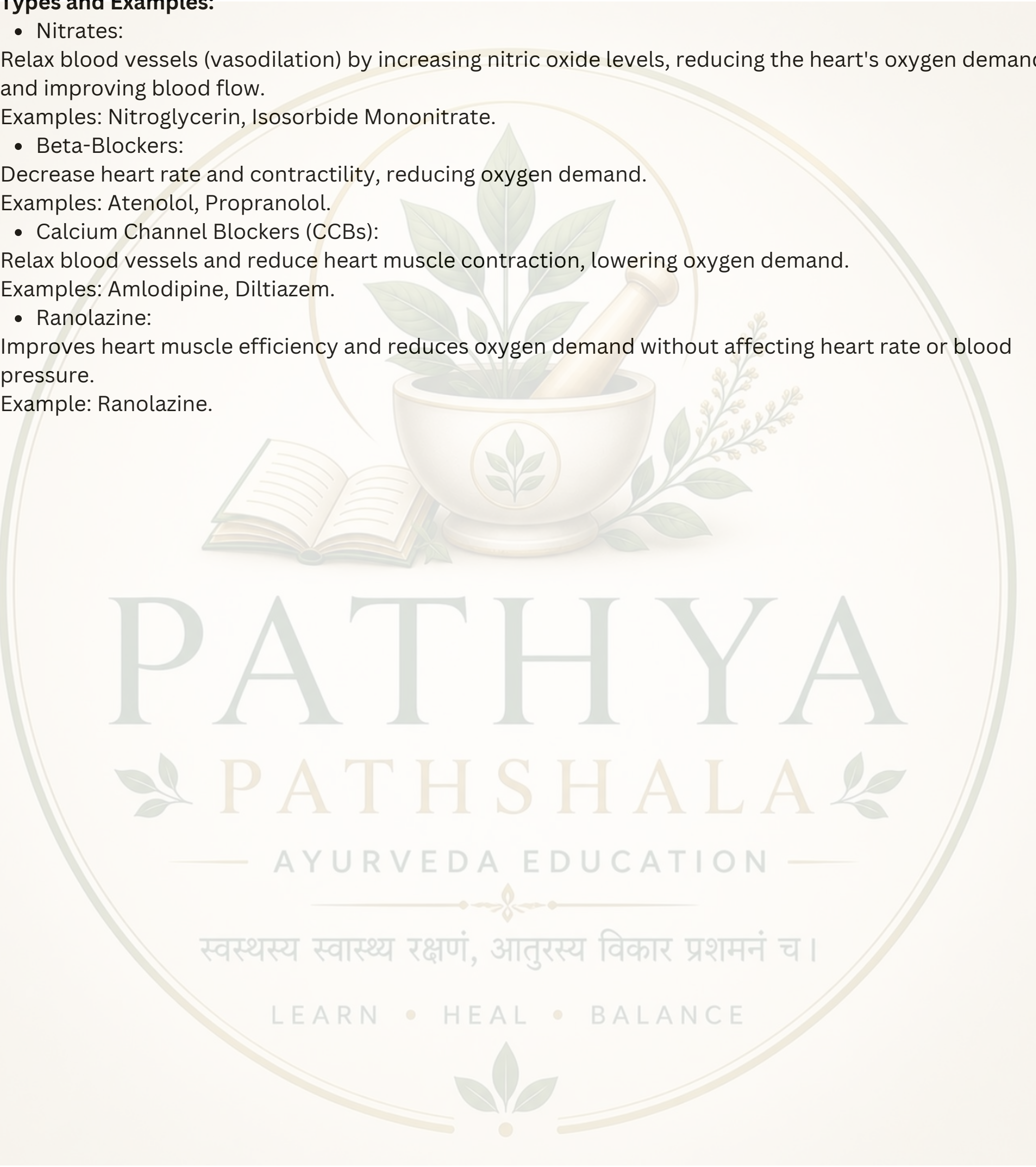
Relax blood vessels and reduce heart muscle contraction, lowering oxygen demand.

Examples: Amlodipine, Diltiazem.

- Ranolazine:

Improves heart muscle efficiency and reduces oxygen demand without affecting heart rate or blood pressure.

Example: Ranolazine.



Define, Describe mode of action & Discuss types with examples of following drugs acting on Kidney as Diuretics

Diuretics

Definition:

Drugs that increase the excretion of water and electrolytes from the body by acting on the kidneys. They are primarily used to treat conditions like hypertension, edema (fluid retention), heart failure, and kidney disorders.

Detailed Mode of Action:

Diuretics work by altering the reabsorption of sodium and water in different parts of the nephron (the functional unit of the kidney).

By inhibiting the reabsorption of sodium, diuretics promote its excretion along with water, reducing blood volume and pressure.

Types and Examples:

Thiazide Diuretics:

Site of Action: Distal convoluted tubule.

Mechanism: Inhibit the sodium-chloride symporter, reducing sodium and water reabsorption.

Uses: Mild to moderate hypertension, mild edema.

Examples: Hydrochlorothiazide, Chlorthalidone, Indapamide.

Loop Diuretics:

Site of Action: Loop of Henle (thick ascending limb).

Mechanism: Inhibit the sodium-potassium-chloride ($\text{Na}^+/\text{K}^+/2\text{Cl}^-$) transporter, causing significant loss of sodium, potassium, and water.

Uses: Severe edema (e.g., in heart failure, liver cirrhosis), acute pulmonary edema.

Examples: Furosemide, Bumetanide, Torsemide.

Potassium-Sparing Diuretics:

Site of Action: Collecting ducts and distal tubule.

Mechanism:

Aldosterone Antagonists: Block aldosterone receptors, reducing sodium reabsorption and potassium excretion.

Example: Spironolactone, Eplerenone.

Epithelial Sodium Channel (ENaC) Inhibitors: Block sodium channels, preventing sodium reabsorption and retaining potassium.

Example: Amiloride, Triamterene.

Uses: Prevent hypokalemia (low potassium), primary hyperaldosteronism.

Carbonic Anhydrase Inhibitors:

Site of Action: Proximal convoluted tubule.

Mechanism: Inhibit carbonic anhydrase, reducing reabsorption of bicarbonate and sodium, leading to increased urine output.

Uses: Glaucoma, altitude sickness, metabolic alkalosis.

Examples: Acetazolamide.

Osmotic Diuretics:

Site of Action: Proximal tubule and loop of Henle.

Mechanism: Increase the osmolarity of the filtrate, pulling water into the nephron and increasing urine output.

Uses: Acute kidney injury, cerebral edema, elevated intraocular pressure.

Examples: Mannitol, Glycerin.

Define, Describe mode of action & Discuss types with examples of following drugs acting as Haematinics, Coagulants, Anticoagulants and Hypolipidaemic Drugs

1. Haematinics

Definition:

Drugs or substances that improve the quality and quantity of blood by increasing hemoglobin levels and red blood cell (RBC) production. These are primarily used to treat anemia.

Detailed Mode of Action:

Haematinics supply essential nutrients required for the synthesis of hemoglobin and RBCs.

Iron Supplements: Provide iron, a crucial component of hemoglobin, improving oxygen-carrying capacity.

Vitamin B12: Essential for DNA synthesis and maturation of RBCs.

Folic Acid: Helps in DNA synthesis and erythropoiesis (formation of RBCs).

Erythropoietin: Stimulates the bone marrow to produce more RBCs.

Types and Examples:

- Iron Preparations:

Oral: Ferrous sulfate, Ferrous gluconate.

Parenteral: Iron sucrose, Ferric carboxymaltose.

- Vitamin B12:

Examples: Cyanocobalamin, Hydroxocobalamin.

- Folic Acid:

Example: Folic acid tablets.

- Erythropoietin Analogues:

Examples: Epoetin alfa, Darbepoetin alfa.

2. Coagulants

Definition:

Drugs that promote blood clotting to prevent or treat excessive bleeding (hemorrhage).

Detailed Mode of Action:

Coagulants act by enhancing the clotting factors or the coagulation process.

Vitamin K: Essential for the synthesis of clotting factors (II, VII, IX, and X) in the liver.

Fibrin Sealants: Promote clot formation locally at the site of bleeding.

Plasma Derivatives: Provide clotting factors directly for replacement therapy.

Types and Examples:

- Vitamin K Preparations:

Examples: Phytomenadione (Vitamin K1), Menadione.

- Fibrin Sealants:

Example: Fibrinogen and thrombin (applied locally).

- Plasma Derivatives:

Example: Fresh frozen plasma, Cryoprecipitate, Factor VIII and IX concentrates.

3. Anticoagulants

Definition:

Drugs that prevent the formation of blood clots or dissolve existing clots to prevent thromboembolic disorders such as stroke, deep vein thrombosis (DVT), and pulmonary embolism.

Detailed Mode of Action:

Anticoagulants interfere with the clotting process by targeting specific clotting factors or enzymes.

- Heparins: Activate antithrombin III, which inhibits clotting factors IIa (thrombin) and Xa.
- Vitamin K Antagonists: Inhibit the synthesis of vitamin K-dependent clotting factors (II, VII, IX, X).
- Direct Oral Anticoagulants (DOACs): Directly inhibit thrombin (factor IIa) or factor Xa.

Types and Examples:

- Heparins:

Examples: Unfractionated Heparin (UFH), Low Molecular Weight Heparins (LMWH) like Enoxaparin, Dalteparin.

- Vitamin K Antagonists:

Example: Warfarin.

- Direct Oral Anticoagulants (DOACs):

Factor Xa inhibitors: Rivaroxaban, Apixaban.

Direct thrombin inhibitors: Dabigatran.

4. Hypolipidaemic Drugs

Definition:

Drugs that lower lipid levels in the blood to prevent or treat hyperlipidemia, a major risk factor for cardiovascular diseases such as atherosclerosis and heart attack.

Detailed Mode of Action:

Hypolipidaemic drugs reduce cholesterol and triglyceride levels by different mechanisms:

- Statins: Inhibit HMG-CoA reductase, the enzyme responsible for cholesterol synthesis in the liver.
- Fibrates: Activate PPAR-alpha receptors, increasing lipid metabolism and reducing triglycerides.
- Bile Acid Sequestrants: Bind bile acids in the intestine, forcing the liver to use more cholesterol to produce bile.
- Cholesterol Absorption Inhibitors: Block dietary cholesterol absorption in the intestine.
- PCSK9 Inhibitors: Increase LDL receptor recycling, enhancing cholesterol removal from the blood.

Types and Examples:

- Statins:

Examples: Atorvastatin, Simvastatin, Rosuvastatin.

- Fibrates:

Examples: Fenofibrate, Gemfibrozil.

- Bile Acid Sequestrants:

Example: Cholestyramine.

- Cholesterol Absorption Inhibitors:

Example: Ezetimibe.

- PCSK9 Inhibitors:

Examples: Alirocumab, Evolocumab.

Define, Describe mode of action & Discuss types with examples of following drugs acting on Gastrointestinal tract as Antacid, Carminatives, Digestants, Antiemetics, Laxatives, Antidiarrhoeal and Hepatoprotective Drug

1. Antacids

Definition:

Drugs that neutralize excess stomach acid (hydrochloric acid) to relieve symptoms of heartburn, indigestion, and peptic ulcers.

Detailed Mode of Action:

Antacids work by neutralizing stomach acid, increasing the pH of gastric contents. This reduces irritation of the stomach lining and esophagus. Some antacids also reduce the activity of pepsin, an enzyme that can contribute to mucosal damage in acidic conditions.

Types and Examples:

Systemic Antacids: Absorbed into the bloodstream.

Example: Sodium bicarbonate.

Non-Systemic Antacids: Not absorbed; act locally in the stomach.

Examples: Magnesium hydroxide (causes diarrhea), Aluminum hydroxide (causes constipation), Calcium carbonate.

Combination Antacids: Blend of magnesium and aluminum hydroxide to balance side effects.

Example: Maalox, Gelusil.

2. Carminatives

Definition:

Drugs that help expel gas from the gastrointestinal tract and relieve bloating and flatulence.

Detailed Mode of Action:

Carminatives act by relaxing the smooth muscles of the gastrointestinal tract and reducing spasms, allowing the gas to be expelled. They may also increase gastric motility to prevent gas accumulation.

Types and Examples:

Natural Carminatives:

Examples: Peppermint oil, Fennel, Ginger.

Pharmaceutical Carminatives:

Examples: Simethicone (reduces surface tension of gas bubbles)

3. Digestants

Definition:

Drugs or enzymes that aid in the digestion of food by supplementing digestive enzymes or enhancing their activity.

Detailed Mode of Action:

Digestants provide enzymes that break down proteins, carbohydrates, and fats in the gastrointestinal tract, improving digestion in conditions like pancreatitis or enzyme deficiencies.

Types and Examples:

Proteolytic Enzymes(for protein digestion):

Examples: Pepsin, Trypsin.

Amylolytic Enzymes(for carbohydrate digestion):

Example: Diastase.

Lipolytic Enzymes(for fat digestion):

Example: Pancrelipase.

4. Antiemetics

Definition:

Drugs that prevent or relieve nausea and vomiting caused by motion sickness, chemotherapy, or gastrointestinal disturbances.

Detailed Mode of Action:

Antiemetics block specific receptors in the brain or gastrointestinal tract involved in the vomiting reflex. These include dopamine, serotonin, histamine, and acetylcholine receptors.

Types and Examples:

Dopamine Antagonists:

Examples: Metoclopramide, Domperidone.

Serotonin (5-HT3) Antagonists:

Examples: Ondansetron, Granisetron.

Antihistamines (H1 Blockers):

Examples: Dimenhydrinate, Meclizine.

Anticholinergics:

Example: Scopolamine.

5. Laxatives

Definition:

Drugs that promote bowel movements to treat constipation.

Detailed Mode of Action:

Laxatives act by softening stools, increasing stool bulk, or stimulating intestinal contractions to facilitate defecation.

Types and Examples:

Bulk-Forming Laxatives: Increase stool bulk by absorbing water.

Examples: Psyllium, Methylcellulose.

Stool Softeners: Soften stools by increasing water penetration.

Example: Docusate sodium.

Osmotic Laxatives: Draw water into the intestine, softening stools.

Examples: Lactulose, Polyethylene glycol.

Stimulant Laxatives: Stimulate bowel contractions.

Examples: Bisacodyl, Senna.

Lubricant Laxatives: Lubricate stool for easier passage.

Example: Mineral oil.

6. Antidiarrhoeal Drugs

Definition:

Drugs that reduce the frequency and severity of diarrhea by slowing intestinal motility or absorbing excess water.

Detailed Mode of Action:

These drugs work by either reducing intestinal motility, absorbing excess water and toxins, or addressing the underlying infection or inflammation.

Types and Examples:

Opioid Derivatives: Slow intestinal motility.

Examples: Loperamide, Diphenoxylate.

Adsorbents: Bind toxins and bacteria in the gut.

Examples: Activated charcoal, Kaolin-pectin.

Antimicrobial Agents: Treat infectious diarrhea.

Example: Metronidazole (for amoebiasis).

Probiotics: Restore normal gut flora.

Example: Lactobacillus.

7. Hepatoprotective Drugs

Definition:

Drugs that protect the liver from damage caused by toxins, infections, or diseases like hepatitis and fatty liver.

Detailed Mode of Action:

Hepatoprotective drugs work by:

Reducing oxidative stress.

Enhancing liver cell regeneration.

Preventing toxin-induced damage.

Types and Examples:

Natural Hepatoprotectives:

Examples: Silymarin (Milk thistle), Liv-52 (Ayurvedic formulation).

Synthetic Hepatoprotectives:

Example: Ursodeoxycholic acid (UDCA) for bile acid regulation.

Antioxidants:

Example: N-Acetylcysteine (NAC).

Define, Describe mode of action & Discuss types with examples of following drugs acting as Antibiotics and Antitubercular.

1. Antibiotics

Definition:

Drugs derived from microorganisms or synthesized chemically that inhibit the growth of or kill bacteria. They are used to treat bacterial infections.

Detailed Mode of Action:

Antibiotics act on specific targets in bacteria to stop their growth or kill them:

Inhibition of Cell Wall Synthesis: Weakens the bacterial cell wall, causing cell lysis.

Inhibition of Protein Synthesis: Prevents the bacteria from making proteins essential for survival.

Inhibition of DNA/RNA Synthesis: Stops bacterial replication by interfering with nucleic acid production.

Disruption of Cell Membrane: Damages bacterial membranes, leading to leakage of cellular contents.

Inhibition of Metabolic Pathways: Targets bacterial metabolic processes like folic acid synthesis.

Types and Examples:

1) Based on Mechanism of Action:

Cell Wall Synthesis Inhibitors:

Examples: Penicillins (Amoxicillin), Cephalosporins (Ceftriaxone), Vancomycin.

Protein Synthesis Inhibitors:

Examples: Tetracyclines (Doxycycline), Macrolides (Azithromycin), Aminoglycosides (Gentamicin).

DNA/RNA Synthesis Inhibitors:

Examples: Fluoroquinolones (Ciprofloxacin), Rifampin.

Cell Membrane Disruptors:

Example: Polymyxins (Polymyxin B).

Metabolic Pathway Inhibitors:

Examples: Sulfonamides (Sulfamethoxazole), Trimethoprim.

2) Based on Spectrum of Activity:

Narrow-Spectrum Antibiotics: Effective against specific bacteria.

Example: Penicillin G (acts on Gram-positive bacteria).

Broad-Spectrum Antibiotics: Effective against a wide range of bacteria.

Example: Amoxicillin-Clavulanic Acid.

3) Based on Source:

Natural: Penicillin (from Penicillium mold).

Semi-Synthetic: Amoxicillin.

Synthetic: Sulfonamides.

2. Antitubercular Drugs

Definition:

Drugs used specifically to treat tuberculosis (TB), caused by *Mycobacterium tuberculosis*.

Detailed Mode of Action:

Antitubercular drugs work by targeting the unique characteristics of *Mycobacterium tuberculosis*:

Inhibition of Cell Wall Synthesis: Blocks the synthesis of mycolic acid, a critical component of the bacterial cell wall.

Disruption of Protein Synthesis: Inhibits the production of essential bacterial proteins.

Inhibition of DNA/RNA Synthesis: Prevents bacterial replication.

Types and Examples:

1) First-Line Antitubercular Drugs:

These are the most effective and are used in the initial treatment phase.

Isoniazid (INH): Inhibits mycolic acid synthesis.

Rifampin: Inhibits bacterial RNA polymerase.

Pyrazinamide: Disrupts bacterial membrane function.

Ethambutol: Inhibits arabinosyl transferase, preventing cell wall synthesis.

Example Combination: HRZE (Isoniazid, Rifampin, Pyrazinamide, Ethambutol).

2) Second-Line Antitubercular Drugs:

Used when there is resistance to first-line drugs or in severe cases.

Fluoroquinolones: Levofloxacin, Moxifloxacin.

Injectables: Amikacin, Capreomycin.

Other Drugs: Ethionamide, Cycloserine.

3) Combination Therapy:

Fixed-Dose Combinations (FDCs): Combine multiple first-line drugs in one pill to ensure compliance.

Example: 4FDC (HRZE in a single tablet).

4) Drugs for Drug-Resistant Tuberculosis (DR-TB):

Multidrug-Resistant TB (MDR-TB): Treated with second-line drugs like Bedaquiline and Linezolid.

Extensively Drug-Resistant TB (XDR-TB): Treated with newer drugs like Pretomanid.

Define, Describe mode of action & Discuss types with examples of following drugs acting as Antifungal, Antiviral, Antimalarial and Anthelmintic Drugs

1. Antifungal Drugs

Definition:

Drugs used to treat infections caused by fungi, including superficial (skin, nails) and systemic fungal infections.

Detailed Mode of Action:

Antifungal drugs act by targeting specific structures or processes in fungi:

Disruption of Cell Membrane: Inhibit synthesis or function of ergosterol (a key component of fungal cell membranes), leading to membrane instability and cell death.

Inhibition of Cell Wall Synthesis: Block the formation of fungal cell walls, weakening the structure.

Inhibition of DNA/RNA Synthesis: Interfere with nucleic acid production, preventing fungal growth.

Types and Examples:

1) Polyenes: Bind to ergosterol and disrupt fungal cell membranes.

Examples: Amphotericin B, Nystatin.

2) Azoles: Inhibit ergosterol synthesis.

Examples: Fluconazole, Ketoconazole, Itraconazole.

3) Echinocandins: Inhibit fungal cell wall synthesis.

Examples: Caspofungin, Micafungin.

4) Allylamines: Block ergosterol synthesis at an earlier step.

Example: Terbinafine.

5) Antimetabolites: Interfere with fungal DNA synthesis.

Example: Flucytosine.

2. Antiviral Drugs

Definition:

Drugs used to treat viral infections by inhibiting viral replication.

Detailed Mode of Action:

Antiviral drugs work by targeting various stages of the viral life cycle:

Inhibition of Viral Entry: Prevent the virus from entering host cells.

Inhibition of Viral Replication: Block enzymes like reverse transcriptase or polymerase required for viral DNA or RNA synthesis.

Inhibition of Viral Assembly and Release: Prevent the formation of mature viral particles or their release from host cells.

Types and Examples:

1) Nucleoside/Nucleotide Analogues: Inhibit viral DNA/RNA synthesis by mimicking building blocks.

Examples: Acyclovir (Herpes), Zidovudine (HIV).

2) Protease Inhibitors: Block viral protein processing.

Examples: Ritonavir, Lopinavir (HIV).

3) Integrase Inhibitors: Prevent viral DNA from integrating into host DNA.

Example: Dolutegravir (HIV).

4) Fusion Inhibitors: Block virus-cell fusion.

Example: Enfuvirtide (HIV).

5) Neuraminidase Inhibitors: Block viral release from infected cells.

Examples: Oseltamivir (Influenza), Zanamivir.

3. Antimalarial Drugs

Definition:

Drugs used to treat and prevent malaria, caused by Plasmodium parasites transmitted through mosquito bites.

Detailed Mode of Action:

Antimalarial drugs target different stages of the parasite’s life cycle:

Blood Schizonticides: Kill parasites in red blood cells, alleviating symptoms.

Liver Schizonticides: Eliminate dormant parasites in the liver to prevent relapse.

Gametocides: Destroy sexual forms of the parasite to prevent transmission.

Types and Examples:

1)Quinolines: Interfere with heme metabolism, leading to parasite death.

Examples: Chloroquine, Quinine, Mefloquine.

2) Antifolates: Inhibit folic acid synthesis, essential for parasite growth.

Examples: Sulfadoxine-Pyrimethamine.

3) Artemisinin Derivatives: Produce free radicals that damage parasites.

Examples: Artemether-Lumefantrine, Artesunate.

4) Atovaquone-Proguanil: Disrupt mitochondrial function and DNA synthesis in parasites.

4. Anthelmintic Drugs

Definition:

Drugs used to treat infections caused by parasitic worms (helminths), including roundworms, tapeworms, and flukes.

Detailed Mode of Action:

Anthelmintic drugs target various structures and functions of helminths:

Paralysis of Worms: Affect the neuromuscular system, causing paralysis and expulsion from the intestine.

Disruption of Energy Metabolism: Interfere with glucose uptake or ATP production, starving the parasite.

Inhibition of Microtubule Synthesis: Prevent cell division and reproduction of worms.

Types and Examples:

1) Benzimidazoles: Inhibit microtubule synthesis.

Examples: Albendazole, Mebendazole.

2) Pyrante Pamoate: Causes spastic paralysis of intestinal worms.

Example: Pyrante.

3) Ivermectin: Binds to chloride channels, causing paralysis.

Example: Ivermectin (used for roundworms and filarial worms).

4) Praziquantel: Increases calcium permeability, leading to paralysis of flukes and tapeworms.

Example: Praziquantel (for schistosomiasis and tapeworms).

5) Diethylcarbamazine (DEC): Inhibits arachidonic acid metabolism, immobilizing worms.

Example: DEC (for filarial infections).

Define, Describe mode of action & Discuss types with examples of following drugs acting on Hormones and Related Drugs as Thyroid Hormone, Thyroid Inhibitors, Insulins, Oral Antidiabetic, Hormonal Contraceptives, Uterine Stimulants and Uterine Relaxants Drugs.

1. Thyroid Hormone Drugs

Definition:

Drugs used to treat hypothyroidism by supplementing or replacing deficient thyroid hormones.

Detailed Mode of Action:

Thyroid hormones regulate metabolism, growth, and energy production:

Synthetic Thyroid Hormones (Levothyroxine): Mimic the action of natural thyroxine (T4), which gets converted to triiodothyronine (T3) in the body. T3 regulates cellular metabolism and protein synthesis.

Types and Examples:

1) Levothyroxine (T4):

Example: Eltroxin, Synthroid.

2) Liothyronine (T3):

Example: Cytomel.

3) Combination Therapy:

Example: Liotrix (T3 + T4).

2. Thyroid Inhibitors

Definition:

Drugs used to treat hyperthyroidism by reducing the production or release of thyroid hormones.

Detailed Mode of Action:

Thioamides: Inhibit thyroid peroxidase, blocking the synthesis of T3 and T4.

Iodides: Inhibit the release of thyroid hormones from the thyroid gland.

Radioactive Iodine: Destroys thyroid tissue selectively by emitting beta radiation.

Types and Examples:

1) Thioamides:

Examples: Methimazole, Propylthiouracil (PTU).

2) Iodides:

Example: Lugol's Iodine.

3) Radioactive Iodine:

Example: Sodium Iodide-131.

3. Insulins

Definition:

Hormones used to manage diabetes mellitus by lowering blood glucose levels.

Detailed Mode of Action:

Insulin binds to receptors on cell membranes, promoting glucose uptake by cells, glycogen storage, and inhibition of gluconeogenesis.

Types and Examples:

1) Rapid-Acting Insulin: Starts working within minutes.

Example: Insulin Lispro, Insulin Aspart.

2) Short-Acting Insulin: Takes effect within 30 minutes.

Example: Regular Insulin.

3) Intermediate-Acting Insulin: Lasts longer but takes time to act.

Example: NPH Insulin.

4) Long-Acting Insulin: Provides a steady glucose-lowering effect.

Example: Insulin Glargine, Insulin Detemir.

.4) Oral Antidiabetic Drugs

Definition:

Medications taken orally to lower blood glucose levels in patients with type 2 diabetes.

Detailed Mode of Action:

Increase Insulin Secretion: Stimulate pancreatic beta cells.

Reduce Insulin Resistance: Improve the body's response to insulin.

Reduce Glucose Absorption: Slow carbohydrate digestion and absorption.

Types and Examples:

1) Sulfonylureas: Stimulate insulin secretion.

Example: Glipizide, Glyburide.

2) Biguanides: Reduce liver glucose production.

Example: Metformin.

3) Thiazolidinediones (TZDs): Improve insulin sensitivity.

Example: Pioglitazone.

4) DPP-4 Inhibitors: Prolong incretin hormone activity.

Example: Sitagliptin.

5) SGLT-2 Inhibitors: Increase glucose excretion through urine.

Example: Dapagliflozin.

5. Hormonal Contraceptives

Definition:

Medications used to prevent pregnancy by altering hormone levels.

Detailed Mode of Action:

Hormonal contraceptives prevent ovulation, thicken cervical mucus, and thin the endometrium to inhibit fertilization and implantation.

Types and Examples:

1) Combined Oral Contraceptives (Estrogen + Progestin):

Example: Ethinylestradiol + Levonorgestrel.

2) Progestin-Only Pills (Mini Pills):

Example: Norethindrone.

3) Injectables:

Example: Medroxyprogesterone Acetate.

4) Implants:

Example: Etonogestrel Implant (Nexplanon).

5) Emergency Contraceptives:

Example: Levonorgestrel (Plan B).

6. Uterine Stimulants

Definition:

Drugs that induce or enhance uterine contractions during labor or to manage postpartum hemorrhage.

Detailed Mode of Action:

Stimulate uterine smooth muscles through receptor activation:

Oxytocin Receptors: Promote uterine contractions.

Prostaglandin Receptors: Induce cervical ripening and contractions.

Types and Examples:

1) Oxytocin: Directly stimulates uterine contractions.

Example: Pitocin.

2) Prostaglandins: Induce cervical ripening and contractions.

Examples: Misoprostol, Dinoprostone.

3) Ergot Alkaloids: Cause sustained uterine contraction.

Example: Ergometrine, Methylergometrine.

7. Uterine Relaxants (Tocolytics)

Definition:

Drugs used to suppress preterm labor by relaxing the uterine muscles.

Detailed Mode of Action:

Beta-2 Agonists: Relax uterine smooth muscles by stimulating beta-2 adrenergic receptors.

Calcium Channel Blockers: Inhibit calcium entry into uterine muscle cells, reducing contractions.

Magnesium Sulfate: Competes with calcium to reduce muscle contraction.

Types and Examples:

1) Beta-2 Agonists:

Example: Terbutaline.

2) Calcium Channel Blockers:

Example: Nifedipine.

3) Prostaglandin Synthesis Inhibitors (NSAIDs):

Example: Indomethacin.

4) Magnesium Sulfate:

Example: Magnesium Sulfate Injection.



Define, Describe mode of action & Discuss types with examples of following drugs acting on as Antiseptics, Disinfectants, Vaccines, Vitamins, Water imbalance and IV

1. Antiseptics

Definition:

Chemical agents applied to living tissues to prevent infection by inhibiting or killing microorganisms.

Detailed Mode of Action:

Disruption of Cell Membranes: Alter the permeability of microbial cell membranes, causing leakage of contents.

Protein Denaturation: Damage microbial proteins and enzymes, rendering them non-functional.

Oxidative Damage: Release reactive oxygen species that kill microbes.

Types and Examples:

Alcohols: Denature proteins and disrupt cell membranes.

Example: Ethanol, Isopropanol.

Chlorhexidine: Disrupts bacterial membranes and precipitates cellular proteins.

Example: Chlorhexidine gluconate.

Iodine Compounds: Oxidize cellular components.

Example: Povidone-Iodine.

Hydrogen Peroxide: Releases reactive oxygen species, damaging microbes.

Example: Hydrogen Peroxide 3%.

2. Disinfectants

Definition:

Chemical agents used on non-living surfaces to kill or reduce microorganisms.

Detailed Mode of Action:

Protein Denaturation: Coagulate or denature microbial proteins.

Lipid Disruption: Destroy lipid membranes of bacteria and viruses.

Enzyme Inhibition: Inactivate microbial enzymes, halting metabolic processes.

Types and Examples:

1) Alcohols:

Example: Isopropanol, Ethanol.

2) Halogens: Oxidize and destroy microbial components.

Example: Sodium Hypochlorite (Bleach).

3) Phenolics: Disrupt cell membranes and denature proteins.

Example: Phenol, Triclosan.

4) Quaternary Ammonium Compounds (QACs): Interfere with microbial membranes.

Example: Benzalkonium Chloride.

3. Vaccines

Definition:

Biological preparations that stimulate the immune system to provide immunity against specific diseases.

Detailed Mode of Action:

Vaccines mimic infection by introducing antigens (weakened or inactive forms of pathogens), which trigger the immune system to produce antibodies and memory cells for future protection.

Types and Examples:

1) Live-Attenuated Vaccines: Contain weakened pathogens.

Example: Measles-Mumps-Rubella (MMR), BCG (Tuberculosis).

2) Inactivated Vaccines: Contain killed pathogens.

Example: Inactivated Polio Vaccine (IPV), Hepatitis A.

3) Subunit/Conjugate Vaccines: Use specific antigens or parts of the pathogen.

Example: Hepatitis B Vaccine, Pneumococcal Vaccine.

4) mRNA Vaccines: Deliver genetic instructions to produce antigens.

Example: COVID-19 Vaccines (Pfizer-BioNTech, Moderna).

5) Toxoid Vaccines: Contain inactivated toxins.

Example: Tetanus, Diphtheria.

4. **Vitamins**

Definition:

Essential organic compounds required in small amounts for normal metabolism, growth, and **overall health**.

Detailed Mode of Action:

Cofactors for Enzymes: Assist enzymes in biochemical reactions.

Antioxidant Action: Protect cells from oxidative damage.

Gene Regulation: Some vitamins regulate gene expression.

Types and Examples:

1) Water-Soluble Vitamins: Dissolve in water and are not stored in the body.

Examples: Vitamin B-complex (B1: Thiamine, B12: Cyanocobalamin), Vitamin C.

2) Fat-Soluble Vitamins: Stored in body fat and require fat for absorption.

Examples: Vitamin A, D, E, K.

5. **Water Imbalance**

Definition:

Disturbance in the body’s water balance caused by dehydration or fluid overload.

Detailed Mode of Action:

Rehydration Therapy: Replenish fluids and electrolytes lost during dehydration.

Diuretics for Fluid Overload: Increase urine output to reduce fluid retention.

Types and Examples:

1) Oral Rehydration Salts (ORS): Restore fluid and electrolyte balance.

Example: ORS solution (Glucose + Sodium + Potassium).

2) Diuretics: Promote urine production.

Example: Furosemide, Hydrochlorothiazide.

6. **Intravenous (IV) Therapy**

Definition:

Administration of fluids, medications, or nutrients directly into the bloodstream via an intravenous line.

Detailed Mode of Action:

Fluid Replacement: Replenish fluids and electrolytes.

Medication Delivery: Ensure rapid and precise delivery of drugs.

Nutritional Support: Provide essential nutrients intravenously when oral intake is inadequate.

Types and Examples:

1) Crystalloids: Contain electrolytes for fluid replacement.

Examples: Normal Saline (0.9% NaCl), Ringer’s Lactate.

2) Colloids: Contain larger molecules for volume expansion.

Examples: Albumin, Dextran.

3) Total Parenteral Nutrition (TPN): Provide all essential nutrients intravenously.

Example: TPN Solution (Amino acids, Dextrose, Lipids).

4) IV Medications:

Examples: Antibiotics (Vancomycin), Pain Relievers (Morphine).

